

Analysis of Avian Biphonic Vocalization Using Computational Modelling

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Abstract

Birds produce complex vocalizations through the coordinated dynamics of neuromuscular control, where the syrinx acts as the sound source, and suprasyringeal vocal tract structures function as acoustic filters. In this study, we developed computational models of the avian syrinx and upper vocal tract, reconstructed from high-resolution micro-CT imaging. Using finite element analysis in COMSOL Multiphysics, we introduce dual sources at the bottom of the trachea to analyze the filtering mechanisms underlying biphonic vocalization. The simulation results are experimentally validated using real bird vocalizations. To further elucidate the biomechanical factors governing resonance modulation, we systematically vary tracheal length, glottal radius, and beak angle. Additionally, we perform an analysis of syllable bandwidth to assess the influence of vocal tract dynamics on the frequency range observed in biphonic vocalizations.

Index Terms: graph convolutional network, data augmentation, mel-spectrogram, multi-label bird classification

1. Introduction

Bird sound is produced by the syrinx [1, 2], a specialized vocal organ located at the junction of the trachea (windpipe) and primary bronchi (air passages to the lungs). During expiration, airflow induces vibrations in its thin membranes, generating the fundamental acoustic signal. The pitch and tone of the produced sound are modulated by the dynamic tension and position of these vibrating membranes. The complexity of avian vocalizations varies across species, depending on the neuromuscular control of the syrinx. Oscine songbirds (e.g., canaries, robins) possess highly developed intrinsic muscles that enable precise control over phonation, allowing for rapid frequency modulations and complex syllable structures. In contrast, suboscines and non-passerine birds (e.g., crows) have a more rigid syrinx with fewer control mechanisms, resulting in less intricate vocal patterns. The primary sound generated by the syrinx undergoes filtering through the suprasyringeal vocal tract, including the trachea, glottis, and beak, shaping the final acoustic output [3–6]. Birds capable of vibrating both sides of the syrinx independently, known as biphonic or bilateral phonators, are primarily songbirds (passerines). This ability allows them to produce two distinct sounds simultaneously, either harmonizing with themselves or generating rapid, complex songs. Notable examples include thrushes, canaries, mockingbirds, lyrebirds, and starlings. Since both sides of their syrinx can function independently, these birds can sing two notes at once, switch frequencies rapidly, or produce continuous sound without interruption. This vocal flexibility results in more intricate and acoustically diverse songs compared to species that utilize only one side of the syrinx. The syrinx anatomy and spectrogram of

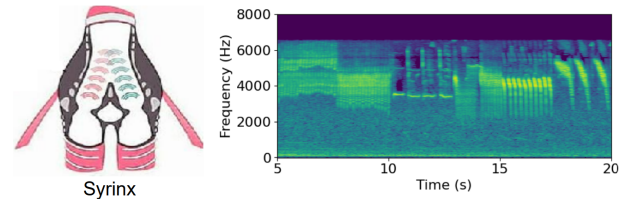


Figure 1: Syrinx and spectrogram of biphonic vocalization

biphonic vocalization are shown in Figure 1.

Computational models can replicate how variations in vocal anatomy produce unique acoustic patterns [7], which serve as species-specific signatures and aid in bioacoustic classification [8–19]. Commonly used electrical network models often neglect the influence of the organs' actual shapes [3, 17]. In [20], researchers introduced a three-dimensional human vocal tract model through two-dimensional finite element analysis. The study [5] examined vocal imitation abilities in parrots and crows compared to humans, utilizing COMSOL Multiphysics to investigate the impact of vocal tract dimensional variations on sound production. The study [21] employed micro-CT imaging and 3D finite element models to simulate sound reception and auditory responses in Little Penguins, predicting their audiometric range from 100 Hz to 10 kHz in both terrestrial and underwater environments. Further, studies such as [22, 23] explored tracheal elongation and formant production in birds, demonstrating that vocal tract geometry significantly influences formant patterns, independent of body size. Two biomechanical models of the syrinx for studying the onset of sound generation and the control of higher harmonics are proposed in [24]. However, the analysis is conducted using a single unilateral pair of labia. In contrast, we analyze the biphonic vocalization using the anatomy of unilateral phonators (birds that vibrate only one side of the syrinx at a time).

We developed a computational model of the syrinx and upper vocal tract, including the trachea, glottis, and beak, which we refer to as the primary model, using COMSOL Multiphysics. Later, we introduced two sources at the bottom of the trachea to investigate the filtering of biphonic vocalization. The response is compared with real-time audio samples of the domestic canary, which has a similar anatomical structure to that of the house sparrow.

2. Experimental Framework

2.1. Computational Modelling

For our computational modelling, we employed X-ray Computed Tomography (CT/microCT) imaging of a whole-body specimen preserved in 70% ethanol and stained with I2KI in an aqueous solution for four weeks. The imaging is performed us-

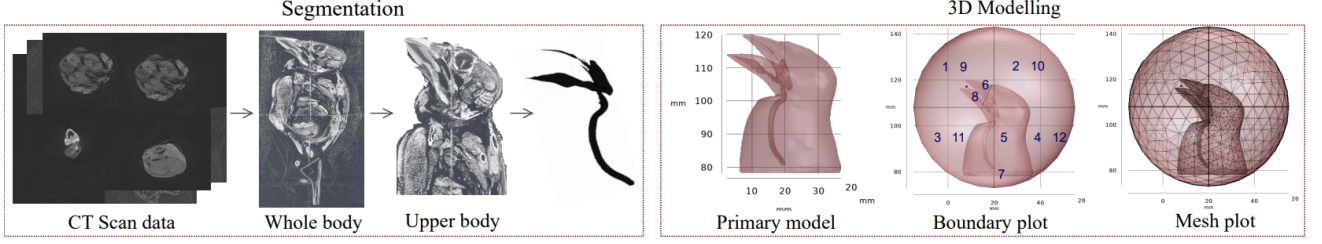


Figure 2: Steps to obtain the primary model (trachea, glottis, and beak), with 12 boundaries marked and mesh plot of the model is given. The sphere surrounding the vocal tract model acts as an artificial boundary to simulate sound propagation into open space.

ing a General Electric Phoenix *v|tome|x* system at the Micro-Computed Tomography Imaging Center (mCTIC) of the National Museum of Natural History, Smithsonian Institution¹. The CT dataset comprised 1,847 slices, with isotropic voxel dimensions of 0.067805 mm along the X, Y, and Z axes. Using Avizo (v. 9.7) and Autodesk Meshmixer, we segmented the CT scans to isolate the upper vocal tract from the surrounding anatomical structures and generated 3D objects.

The 3D models are imported into COMSOL Multiphysics [25] to build the primary model, featuring a 30.58 mm trachea, beak angles of 10° or 20°, and a 0.7 mm glottis radius [26]. A virtual joint is integrated into the lower beak using Blender software, facilitating the desired adjustments. To modify the glottis radius, a cylindrical insert is introduced into the larynx, reducing its surface area and creating a sharp-edged interface at the junction of the glottis and cylinder. Comparative analysis between smooth and sharp-edged glottis geometries, conducted using COMSOL's *differences tool for end corrections, revealed minimal impact. The tongue is excluded, as [26] found its position had no effect on total sound pressure level (SPL). Figure 2 outlines the model generation workflow.

Modelling Interface: The COMSOL Multiphysics Pressure Acoustics, Frequency Domain interface is used to model the pressure waves in a fluid based on the inhomogeneous Helmholtz equation [27]. The pressure wave propagation, accounting for contributions from both monopole and dipole acoustic sources, is mathematically described by:

$$\nabla \cdot \left(-\frac{1}{\rho_c} (\nabla p_t - \mathbf{q}_d) \right) - \frac{k_{eq}^2 p_t}{\rho_c} = Q_m \quad (1)$$

where p_t is the total pressure (Pa), and the wave number $k_{eq} = \frac{\omega}{c}$, with $\omega = 2\pi f$ is the angular frequency for a given frequency f (Hz). The complex-valued parameters ρ_c and c_c represent density (kg/m³) and speed of sound (m/s) in the computational model. The terms $\mathbf{q}_d$ (N/m³) and Q_m (1/s²) represent dipole and monopole domain sources, respectively.

Few assumptions have been considered for model formation. The medium is assumed to be lossless, with no thermal conduction or viscosity. The syrinx output is modelled as a pressure source at the boundary with constant acoustic pressure, $p_t = p_0$. Total pressure, p_t is the sum of background pressure p_b and scattered field p_s . Here $p_b = 0$, then $p_t = p_s = p$. The boundary conditions are applied to ensure accurate acoustic simulations and minimize reflections [28].

Sound-hard Boundary Condition

$$-n \cdot \left(-\frac{1}{\rho_c} (\nabla p_t - \mathbf{q}_d) \right) = 0; \quad \frac{\partial p_t}{\partial n} = 0 \quad (2)$$

where n is a vector that is perpendicular (normal) to the surface at every point on the boundary.

¹<https://www.morphosource.org/>

Impedance Boundary Condition

$$n \cdot \left(-\frac{1}{\rho_c} (\nabla p_t - \mathbf{q}_d) \right) - \frac{i\omega p_t}{Z} = 0 \quad (3)$$

where Z is the acoustic input impedance (Pa·s/m). For skin impedance, physiological impedance model (human skin) is used and the impedance boundary condition in equation (4) uses,

$$Z = i\omega L + R + \frac{1}{i\omega C} \quad (4)$$

$$R = 6 \times 10^4 \frac{\text{kg}}{\text{m}^2 \text{s}}, \quad C = 7.95 \times 10^{-10} \frac{\text{m}^2 \text{s}^2}{\text{kg}}, \quad L = 3.53 \frac{\text{kg}}{\text{m}^2}.$$

The bone impedance and air impedance are user-defined, and the boundary condition in equation (4) uses:

$$Z(\text{bone impedance}) = 1850 [\text{kg/m}^3] \cdot 3000 [\text{m/s}],$$

$$Z(\text{air impedance}) = 1.2 [\text{kg/m}^3] \cdot 343 [\text{m/s}]$$

Radiation Boundary Condition

$$\begin{aligned} & -n \cdot \left(-\frac{1}{\rho_c} (\nabla p_t - \mathbf{q}_d) \right) + \left(ik_{eq} + \frac{1}{r} \right) \frac{p}{\rho_c} - \frac{r \Delta_{\parallel} p}{2\rho_c (ik_{eq} r + 1)} \\ & = -\frac{r \Delta_{\parallel} p_i}{2\rho_{0c} (ik_{eq} r + 1)} + \left(ik_{eq} + \frac{1}{r} \right) \frac{p_i}{\rho_c} + n \cdot \frac{1}{\rho_c} \nabla p_i \end{aligned} \quad (5)$$

The parameter configurations for physical modelling is shown in Table 1 and those boundaries are marked in Figure 2. The reference sound pressure for air is set at 20 μPa , and SPL is consistently measured at a fixed distance in the middle of the beak.

Table 1: Acoustic boundary conditions

Name	Level [Selection]
Pressure Acoustics	Domain [All]
Sound Hard Boundary (Wall)	Boundary [All]
Initial Values	Domain [All]
Exterior Field Calculation	Boundary [1-4, 9-12]
Spherical Wave Radiation	Boundary [1-4, 9-12]
Pressure	Boundary [7]
Air Impedance	Boundary
Skin Impedance	Boundary [5]
Bone Impedance	Boundary [6,8]

A user-controlled mesh with a custom element size is used [29]. The maximum mesh size is calculated as $\frac{c_0}{f_{\max}}/5$, where c_0 is the speed of sound and $f_{\max}$ is the maximum frequency, resulting in a value of 0.00885 m.

2.2. Validation using Real Bird Vocalizations

To investigate the relationship between upper vocal tract geometry and sound filtering in birds, we analyzed video data

of house sparrows and domestic canaries. House sparrow (*Passer domesticus*) vocalizations [30] consists of short, varied 'chirps' produced at a high repetition rate, while domestic canaries (*Serinus canaria*) exhibits biphonic vocalization produce melodious, continuous songs with rich harmonic structures. The video recordings of house sparrow chirping calls (30 seconds) and domestic canary bird calls (15 seconds) are taken with an audio sampling rate of 16 kHz and were sourced from <https://mbirdtraining.com/>. A 1.5 second segment of the audio is extracted and displayed on a computer monitor as individual frames synchronized with the corresponding sound track. Frames are extracted from the video using OpenCV's `cv2.VideoCapture`. Figure 4 (a) presents video frames illustrating the birds' beak movements during vocalization. Each frame captures a slightly different beak angle, showing the motion over time. Specifically, frames A and B correspond to beak angles of 10° and 20° , respectively. Figure 4 (b) displays the amplitude plot of the vocalizations. Similarly, the analysis is conducted for domestic canary vocalizations.

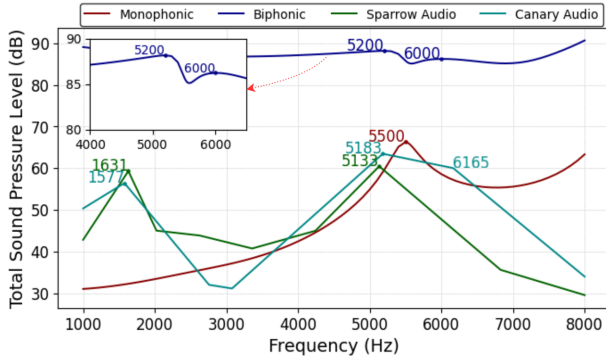


Figure 3: SPL for monophonic and biphonic models with 10° beak angle, along with formants from sparrow (unilateral phonator) and canary (bilateral phonator) audio samples

2.3. Implementation

In this study, we modelled the bird's vocal tract resonance using COMSOL Multiphysics' pressure acoustics frequency domain interface over the 1-8 kHz range typical of house sparrow vocalizations [31]. To investigate biphonic sound production and modulation, we employed two sources, with the tracheal input pressure (syrinx output) set to 1 Pa (94 dB). The simulations are validated using real bird vocalizations. We also modified the computational models to analyze the effects of tracheal length, glottal radius, and beak angle on resonance. Additionally, syllable bandwidth analysis is performed to assess how variations in vocal tract flexibility influence the frequency range in biphonic sound production.

3. Results and Discussion

3.1. Sound Filtering in Biphonic Vocalization

Figure 3 highlights the role of biphonic sound production and filtering through simulation and experimental validation with a 10° beak angle. The primary model exhibits distinct resonance patterns, with the monophonic model (red) showing a single resonant frequency at 5.5 kHz, while the biphonic model (blue) generates two resonant frequencies at 5.2 kHz and 6 kHz. This suggests that the trachea, beak, and associated structures contribute to resonance in the mid-frequency range. To experimentally validate this, we conducted a detailed analysis of house

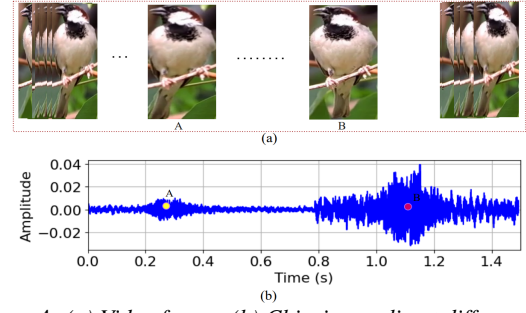


Figure 4: (a) Video frames (b) Chirping audio at different beak angles of house sparrow

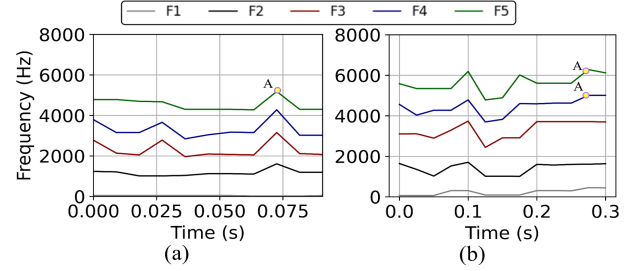


Figure 5: Formant frequencies (a) House sparrow (b) Domestic canary at $A=10^\circ$ beak angle

sparrow and domestic canary vocalizations. Figure 5 illustrates the formant frequencies over time at a 10° beak angle, showing a single formant at 5.13 kHz for the house sparrow (Figure 5a) and two distinct formants at 5.18 kHz and 6.16 kHz for the domestic canary (Figure 5b). These findings indicate that the house sparrow exhibits a single resonant frequency (green), while the domestic canary (cyan) produces dual resonant frequencies, as shown in Figure 3. This confirms the domestic canary's ability to produce and modulate biphonic sounds. For 20° beak, graphs are provided as supplementary material².

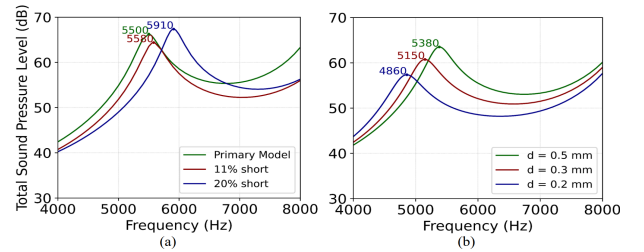


Figure 6: SPL for monophonic model varying (a) tracheal length (b) radius

3.2. Morphological Geometry and Sound Filtering

We modified the 3D geometry to examine the influence of tracheal length, glottis radius, and beak angle on monophonic and biphonic sound production. Previous studies reported tracheal elongation of 3% in zebra finches [32] and 11% in white house sparrows [33] during vocalization, causing minimal frequency shifts. To assess the effects of tracheal length, we reduced the initial tracheal length by 11% and 20% while maintaining a consistent diameter of approximately 1 mm. The simulation results for the monophonic model, shown in Figure 6, indicate a shift in resonance frequency from 5.5 kHz to 5.58 kHz and 5.91 kHz for the 11% and 20% shorter models, respectively. Similarly,

²Additional graphs are provided as a supplementary material

the simulation results for the biphonic model, shown in Figure 7, exhibit two resonant peaks. The first peak shifts from 5.2 kHz to 5.3 kHz and 5.6 kHz for the 11% and 20% shorter models, respectively, while the second peak shifts from 6 kHz to 6.3 kHz for the 20% shorter model in tracheal length, demonstrating that resonance frequency increases with decreasing tube length, though anatomical influences introduce nonlinear effects [33].

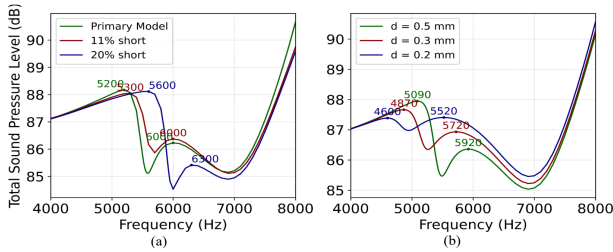


Figure 7: SPL for biphonic model varying (a) tracheal length (b) radius

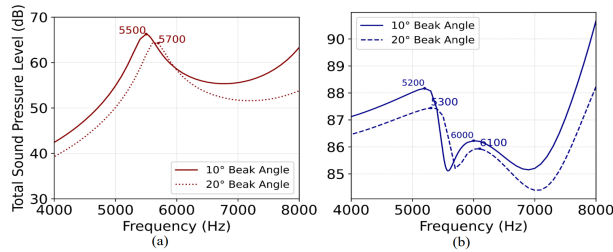


Figure 8: SPL for varying beak (a) monophonic (b) biphonic

Although studies on the larynx are limited, it is known that laryngeal muscle contraction reduces vocal tract diameter [34]. To simulate this effect, we varied the glottal radius and analyzed its impact on frequency. Figure 6, 7 illustrate the variation in resonance with glottis radius for monophonic and biphonic models. In the monophonic model, a glottis with a 0.5 mm radius exhibits an increase of 0.52 kHz compared to a 0.2 mm radius, as shown in Figure 6. For the biphonic model, which features two resonant peaks, a glottis with a 0.5 mm radius shows an increase of 0.49 kHz compared to a 0.2 mm radius for the first peak, and an increase of 0.40 kHz for the second peak, as shown in Figure 7. This indicates that an increase in glottis radius leads to a rise in resonance frequency.

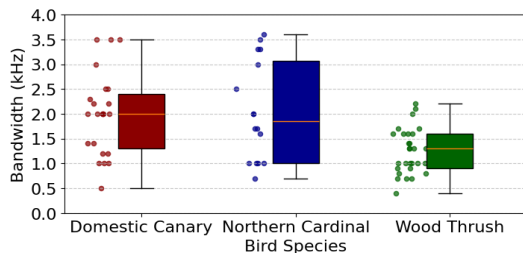


Figure 9: Syllable bandwidth distribution of three biphonic vocalizations

In Figure 8, resonant peaks shift to higher frequencies as the beak angle increases from 10° to 20°. For the monophonic model, the resonance frequency rises from 5.5 kHz at a 10° angle to 5.7 kHz at 20°. Similarly, in the biphonic model, the first peak increases from 5.2 kHz to 5.3 kHz at a 10° angle,

while the second peak increases from 6 kHz to 6.1 kHz at a 20° angle. This frequency shift indicates that a wider beak angle increases the vocal tract's opening, thereby altering its resonant frequencies. The reduced length of the effective resonating tube favors the transmission of higher frequencies [35].

3.3. Bandwidth of Syllables in Biphonic Vocalization

A syllable is the smallest unit of a bird's song, typically a single continuous sound produced within one breath cycle. The arrangement of these syllables into a structured pattern forms a phrase. In biphonic vocalization, a syllable contains two independent frequency components, while a phrase consists of a sequence of syllables, each of which may be biphonic. Biphonic vocalization often results in sideband frequencies and subharmonics. Three birds, namely the domestic canary, wood thrush, and northern cardinal, are selected from the Xeno-canto database [36]. For each species, 10 files with variable durations are used to analyze the bandwidth of syllables. Figure 9 shows the bandwidth distribution and IQR (orange lines) across three bird species, highlighting vocal tract flexibility and biphonic syllable production. The wood thrush exhibits the narrowest bandwidth range (0.4–2.2 kHz, median 1.3 kHz) with the smallest IQR of 0.7 kHz, indicating a more constrained vocal tract. Both the domestic canary (0.5–3.5 kHz, IQR 1.1 kHz, median 2.2 kHz) and the northern cardinal (0.7–3.6 kHz, IQR 1.8 kHz, median 2.5 kHz) demonstrate broader bandwidth ranges and greater variability, suggesting more adaptable vocal tract configurations. Their overlapping frequency ranges indicate high vocal flexibility, enabling them to produce a variety of sounds, including biphonic vocalizations [37].

3.4. Insights and Future Scope

The literature reports that different articulated parts of the oscine vocal tract, including trachea, glottis, oropharyngeal cavity, and beak, functions as an integrated system to shape vocalizations. Our model demonstrates how morphological changes influence vocal tract resonance. By extending this framework, we can synthesize species-specific bird calls, enabling comparative analysis of vocal adaptation across taxa. Further, our model though based on unilateral phonator anatomy, offers valuable insights into bilateral phonation, where birds generate harmonically complex sounds using independent sources. This opens pathways for tonality studies, where harmonic decay patterns can quantify species' vocal characteristics. Future research could integrate real-time vocal tract deformation, allowing the generation of synthetic bird calls that closely resemble natural vocalization. Additionally, cross-species modeling could help investigate evolutionary constraints on vocal tract morphology, shedding light on how birds optimize their acoustic signals for communication, mating, and territorial behavior.

4. Conclusion

Our study demonstrates that real domestic canary vocalizations exhibit two resonant peaks in the mid-frequency range, aligning with the simulation results of our biphonic computational model. We examined how variations in tracheal length, glottal radius, and beak angle influence resonance, finding that shorter tracheal lengths and wider beak angles favor higher frequencies. Bandwidth analysis further reveals that the northern cardinal and domestic canary possess the most flexible vocal tracts, allowing them to produce a wider range of frequencies and more complex vocalizations, including biphonic sounds. These findings underscore the role of vocal tract morphology in shaping avian vocalizations.

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